instead, words within a distance of 2-3 characters are the most likely anyway).

As you can see in figure 6.1, the smallest distance is obtained for the path “ant”: there is, however, a catch! This trie, in fact, doesn’t contain “`ant`” as a key, and therefore we can’t take this value into consideration.

There are, instead, several keys at distance 2, and any of them, or all of them, can be returned as a suggestion.

How fast can we find a suggestion? As we know at this point, after the discussion in section 6.1, searching a string in a trie has a better worst-case running time than the alternatives: `O(m)` comparisons for a string of length `m`, while for hash tables or binary search trees it would be `O(m + log(n))` at best.

6.4.2 Strings Similarity

The similarity between two strings is a measure of the distance that separates them. Usually, it's some function of the number of changes needed to transform one string into the other.

Two examples of these distances are:

- The *Levenshtein distance*, the number of single-character edits;
- The *Hamming distance*, the number of positions in which the strings are different.